

Optimising E-Commerce Shipping Strategy Performance

Introduction

In the competitive e-commerce market, shipping strategy is crucial in shaping customer buying behaviour and overall business performance. Delivery options affect customer satisfaction, purchasing decisions, and spending levels. Therefore, management must evaluate which shipping methods offer the most value to the organisation. This report analyzes customer shopping behaviour using data visualization techniques to compare shipping types based on customer participation, average spending, and estimated revenue. The goal of the analysis is to support decisions by identifying the shipping methods that strike the best balance between customer popularity and financial contribution.

Findings

The analysis shows clear differences in performance among shipping methods. The customer distribution chart reveals that one shipping type attracts the most customers, indicating strong popularity and accessibility. However, the average spending visualisation shows that a higher customer volume does not always lead to higher spending. Some shipping options attract fewer customers but generate greater average purchase values. This suggests that these options appeal to customers willing to pay more for faster or more convenient delivery. The estimated revenue analysis combines customer count and average spend, giving a more balanced view of business value. The results indicate that overall performance depends on both customer participation and spending levels, not just popularity. The comparative visuals reveal that some shipping methods perform well through volume, while others excel with higher-value transactions. This highlights the need for management to consider various performance indicators when assessing the effectiveness of shipping strategies.

Evaluation of Visualisation



Figure 1: Customer Count by Shipping Type

This bar chart illustrates the number of customers selecting each shipping method, allowing clear identification of the most popular delivery option. A bar chart was chosen because it is the most effective visual for comparing categorical frequencies. It enables quick comparison across shipping types and helps management understand customer preference patterns and demand levels.

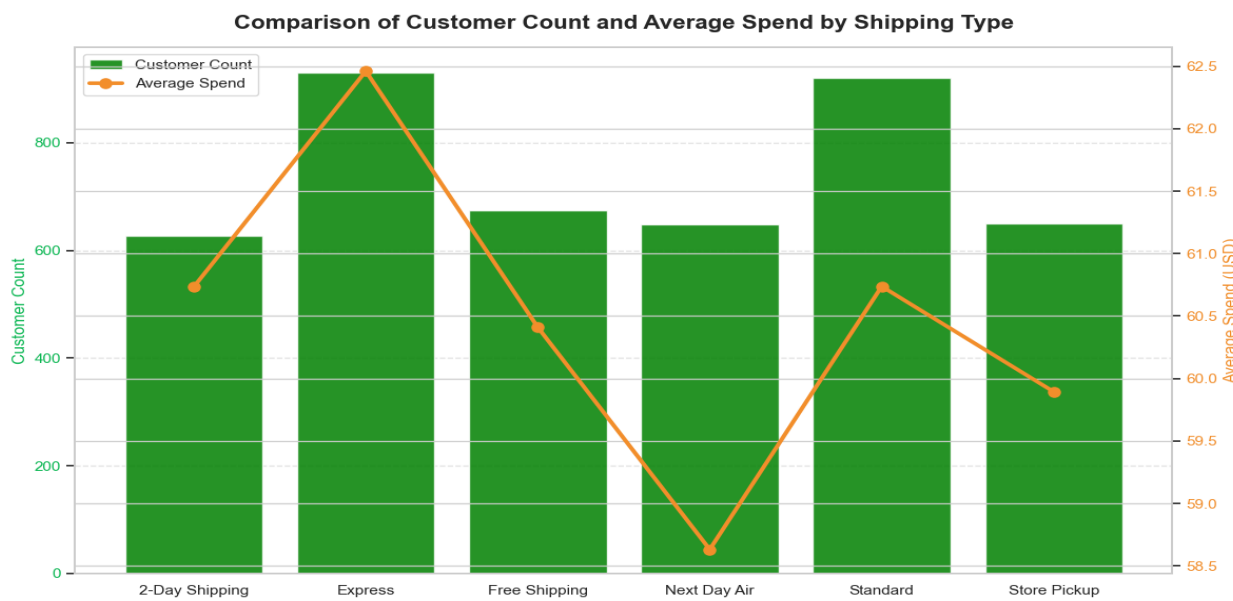


Figure 2: Dual Axis Shipping Performance Chart

This dual-axis chart combines customer count and average spend into one visual to compare popularity and spending behaviour simultaneously. It helps reveal whether highly popular shipping methods also generate strong customer spending. However, dual-axis charts must be interpreted carefully since different scales may visually exaggerate relationships.

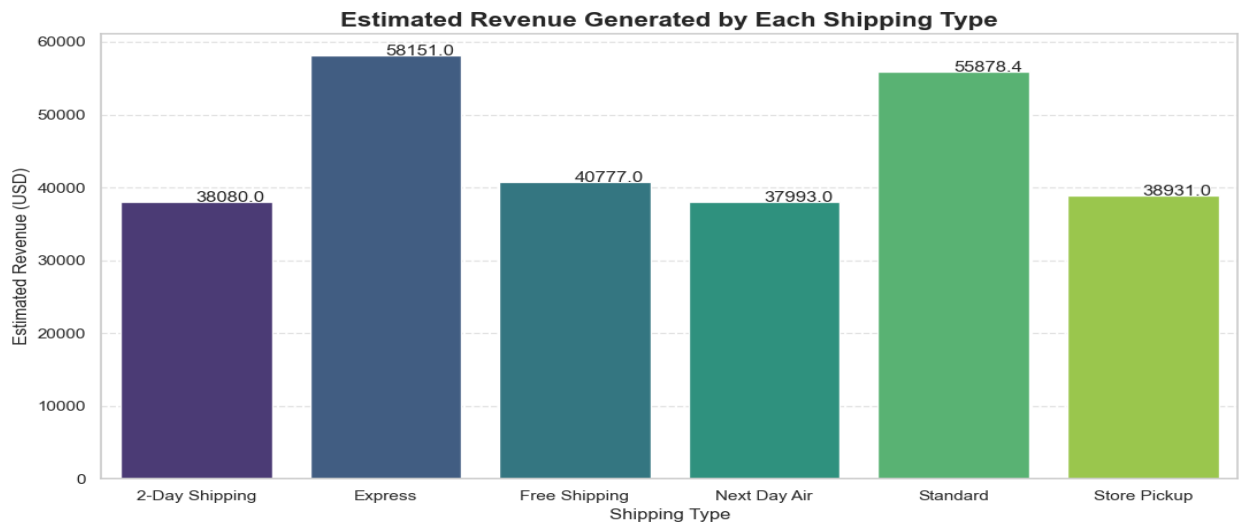


Figure 3: Estimated Revenue Generated by Each Shipping Type

This visual presents estimated revenue calculated using customer count multiplied by average spend. The bar chart format was chosen because it clearly ranks shipping methods according to overall business value. By combining demand and spending power into one metric, this chart directly supports strategic decision-making and prioritisation of shipping investments.



Figure 4: Customer Behaviour Scatter Plot

The scatter plot explores the relationship between customer purchasing behaviour and revenue contribution. This visual was selected because scatter plots are effective for identifying patterns, clusters, and variability within individual customer data rather than aggregated summaries. It helps determine whether repeat purchasing behaviour is associated with higher revenue outcomes and supports deeper behavioural analysis.

Recommendation

Based on the visual analysis, management should focus on two shipping methods that together maximize both customer reach and financial value. The first priority should be **Standard Shipping**, as it attracts the largest number of customers and plays a key role in maintaining steady transaction volumes. Promoting this option through marketing incentives or loyalty programs can enhance customer retention and support stable revenue generation. The second priority should be **Express Shipping**, which shows higher average customer spending despite lower overall usage. This indicates strong revenue potential if operational improvements boost adoption. Investment in service efficiency, delivery reliability, or pricing optimisation might make it more appealing to customers.